

SOLVED EXAMPLE

Q. A coil of 200 turns is wound uniformly over a wooden ring having a mean circumference of 600 mm and a uniform cross-sectional area of 500 mm². If the current through the coil is 4.0 A, calculate (a) the magnetic field strength; (b) the flux density; (c) the total flux.

Ans: $N = 200$, $I = 4 \text{ A}$, $l = 600 \text{ mm} = 0.6 \text{ m}$, $A = 500 \text{ mm}^2 = 5 \times 10^{-4} \text{ m}^2$

(a) $H = N.I / l = 200 \times 4 / 0.6 = 1333.3 \text{ A/m}$

(b) $B = \mu_0 H = 4\pi \times 10^{-7} \times 1333.3 = 1.6755 \times 10^{-3} \text{ T} = 1.6755 \text{ mT}$

(c) $\phi = B.A = 1.6755 \times 10^{-3} \times 5 \times 10^{-4} = 8.376 \times 10^{-7} \text{ Wb} = 0.8376 \text{ } \mu\text{Wb}$

SOLVED EXAMPLE

Q. Calculate the magnetomotive force required to produce a flux of 0.015 Wb across an air-gap 2.5 mm long, having an effective area of 200 cm².

Ans: $\phi = 0.015 \text{ Wb}$, $l = 2.5 \text{ mm} = 2.5 \times 10^{-3} \text{ m}$, $A = 200 \text{ cm}^2 = 2 \times 10^{-2} \text{ m}^2$

$$B = \phi / A = 0.015 / (2 \times 10^{-2}) = 0.75 \text{ T}$$

$$H = B / \mu_0 = 0.75 / (4\pi \times 10^{-7}) = 5.97 \times 10^5 \text{ A/m}$$

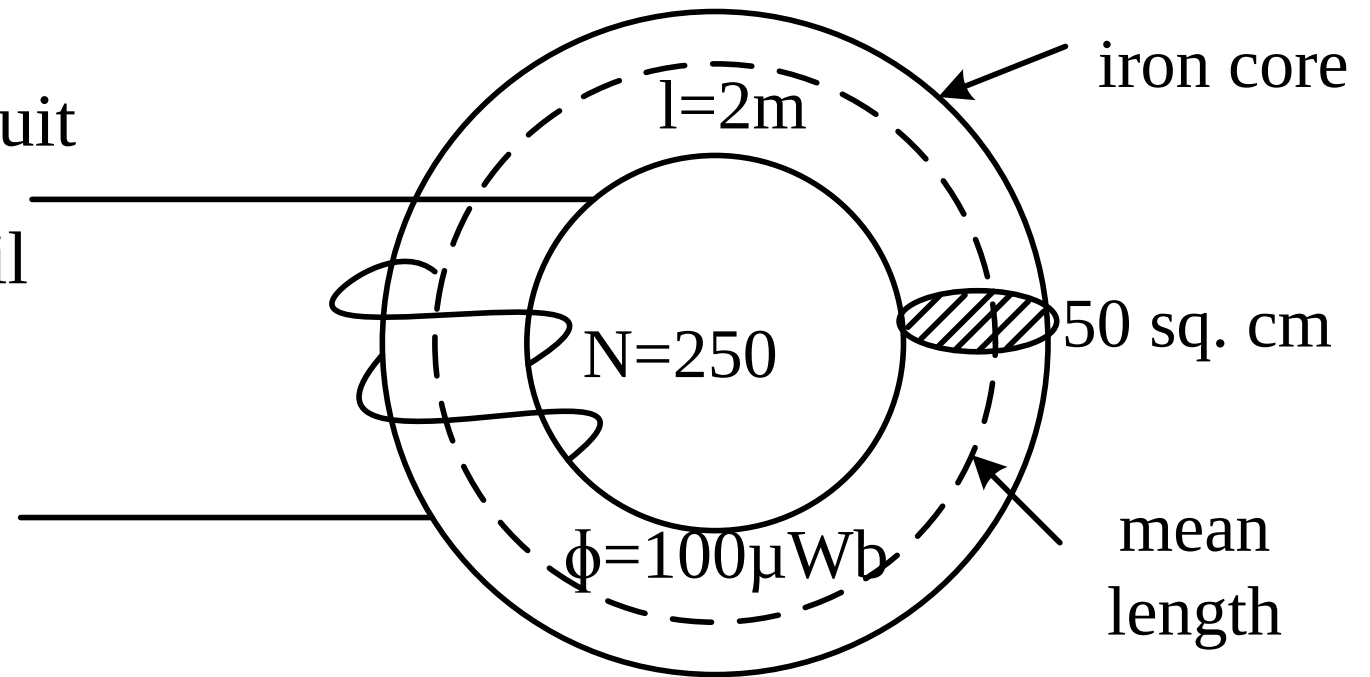
$$F (\text{mmf}) = H.l = 5.97 \times 10^5 \times 2.5 \times 10^{-3} = 1.49 \times 10^3 \text{ A}$$

SOLVED EXAMPLE

Q. The simple magnetic circuit shown has a cross-sectional area of 50 cm^2 and mean length of 2 m . The relative permeability of the core is 100 . The coil has 250 turns and the flux produced is $100 \mu\text{Wb}$.

Find:

- (a) Reluctance of the magnetic circuit
- (b) Current flowing through the coil



Ans:

(a) Reluctance, $S = \frac{l}{\mu A}$

$$\mu = \mu_0 \mu_r = 4\pi * 10^{-7} * 100 = 4\pi * 10^{-5}$$

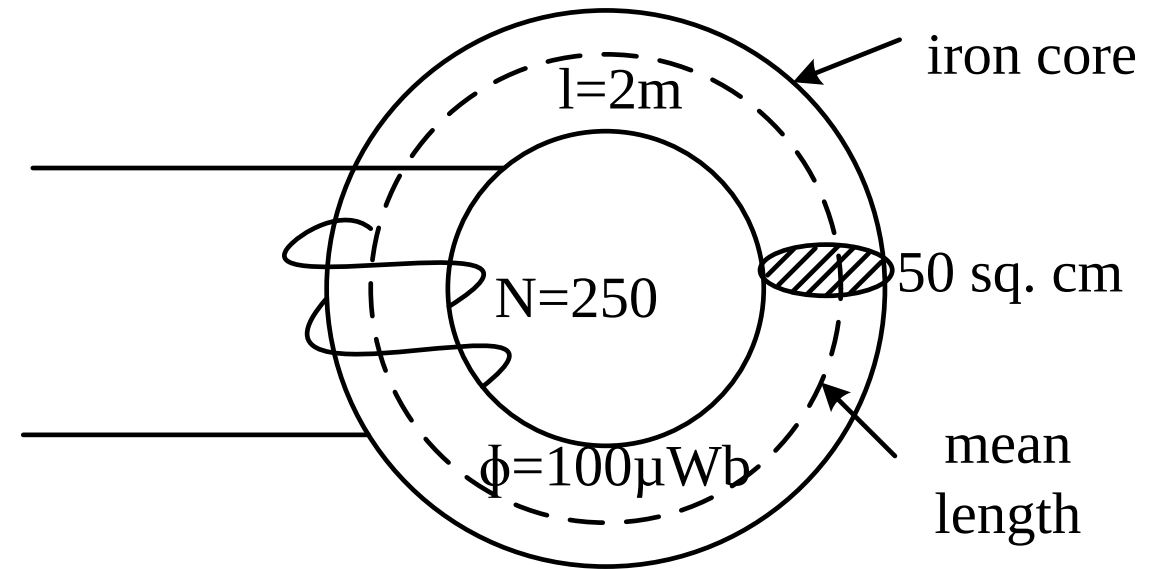
$$S = \frac{2}{4\pi * 10^{-5} * 50 * 10^{-4}} = 3.18 * 10^6 \text{ A/Wb}$$

(b) Magnetic field strength, $H = \frac{mmf}{l} = \frac{NI}{l}$

$$I = \frac{Hl}{N}$$

$$H = \frac{B}{\mu} = \frac{\phi}{A\mu} = \frac{100 * 10^{-6}}{50 * 10^{-4} * 4\pi * 10^{-5}} = 159.15 \text{ A/m}$$

$$I = \frac{Hl}{N} = \frac{159.15 * 2}{250} = 1.27 \text{ A}$$



SOLVED EXAMPLE

Q. The air gap in a magnetic circuit is 1.5 mm long and 2500 mm² in cross-sectional area. Calculate (a) The reluctance of the air gap (b) The *mmf* required to set up a flux of 800 μWb in the air gap.

Ans: (a) Reluctance, $S = \frac{l}{\mu A}$

$$\mu = \mu_0 \mu_r = 4\pi * 10^{-7} * 1 = 4\pi * 10^{-7} \quad (\mu_r \text{ of air} = 1)$$

$$S = \frac{1.5 * 10^{-3}}{4\pi * 10^{-7} * 2500 * 10^{-6}} = 4.77 * 10^5 \text{ A/Wb}$$

$$(b) H = \frac{B}{\mu} = \frac{\phi}{A\mu} = \frac{800 * 10^{-6}}{2500 * 10^{-6} * 4\pi * 10^{-7}} = 2.55 * 10^5 \text{ A/m}$$

$$mmf = Hl = 2.55 * 10^5 * 1.5 * 10^{-3} = 382.5 \text{ A}$$

MAGNETIC CIRCUIT ANALOGY

WITH ELECTRICAL CIRCUIT

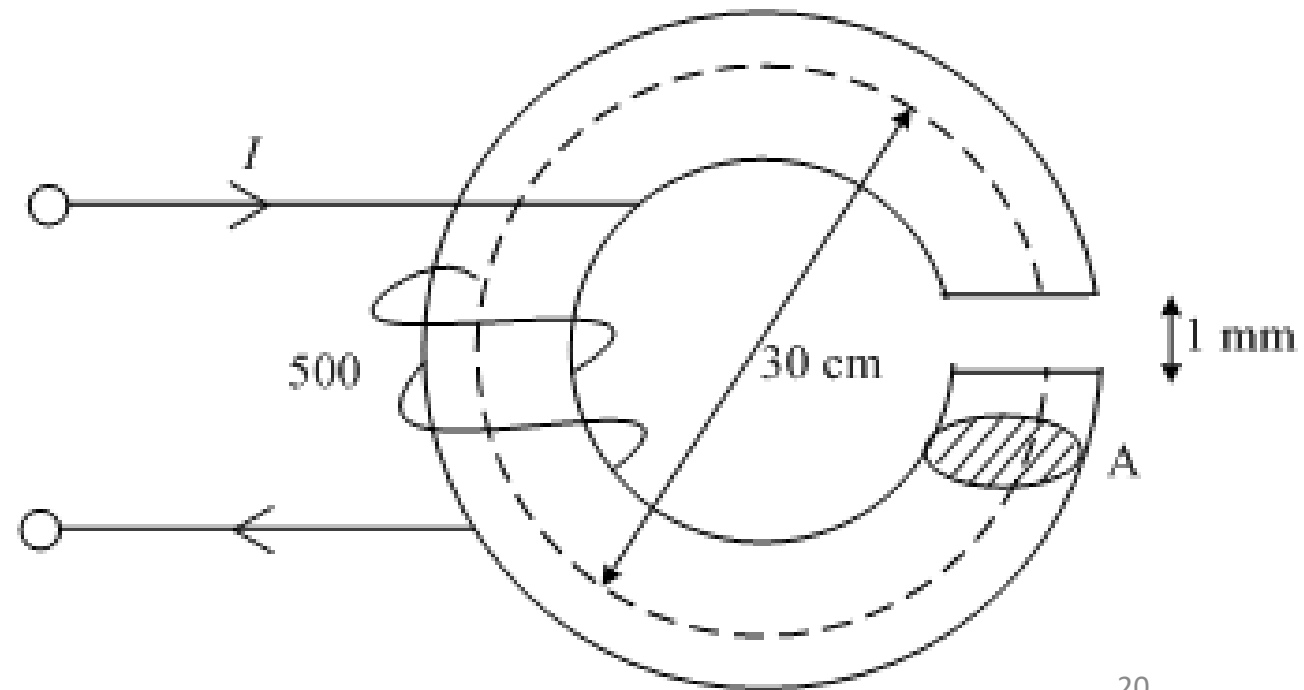
S.No	Magnetic circuit quantity	Electrical circuit quantity
1	Magnetic flux density $\mathbf{B}$ (T = Wb/m ²)	Current density $\mathbf{J}$ (A/m ²)
2	Magnetic flux ϕ (Wb)	Current $\mathbf{I}$ (A)
3	Magnetic field intensity $\mathbf{H}$ (A/m)	Electric field intensity $\mathbf{E}$ (V/m)
4	m.m.f. $\mathbf{F}$ (A)	e.m.f. $\mathbf{E}$ (V)
5	Reluctance $\mathbf{S}$ (A/Wb)	Resistance $\mathbf{R}$ (Ω = V/A)
6	Permeance (H = Wb/A)	Conductance (S = A/V)
7	Permeability μ (H/m)	Conductivity σ (S/m)

SOLVED EXAMPLE

Q. A ring of 30-cm mean diameter is made using a cylindrical iron rod of diameter 2.5 cm. A saw-cut 1-mm wide is made through the ring to create an air-gap. A coil with 500 turns of wire is wound on the ring. Calculate the current required in the exciting coil to produce a flux of 4 mWb in the ring. Assume the relative permeability of iron at this flux density as 800. Neglect any leakage or fringing of the magnetic field.

Ans: $A = \pi r^2 = 3.14 \times (0.025 \text{ m})^2$
 $= 4.91 \times 10^{-4} \text{ m}^2$

$$\begin{aligned}\Rightarrow B &= \phi / A \\ &= (4 \times 10^{-3} \text{ Wb}) / (4.91 \times 10^{-4} \text{ m}^2) \\ &= 8.15 \text{ Wb/m}^2\end{aligned}$$



First consider air gap:

$$H_1 = B / \mu_0 = 8.15 / (4\pi \times 10^{-7}) \text{ A/m}$$
$$= 6.49 \times 10^6 \text{ A/m}$$

$$l_1 = 1 \text{ mm} = 10^{-3} \text{ m}$$

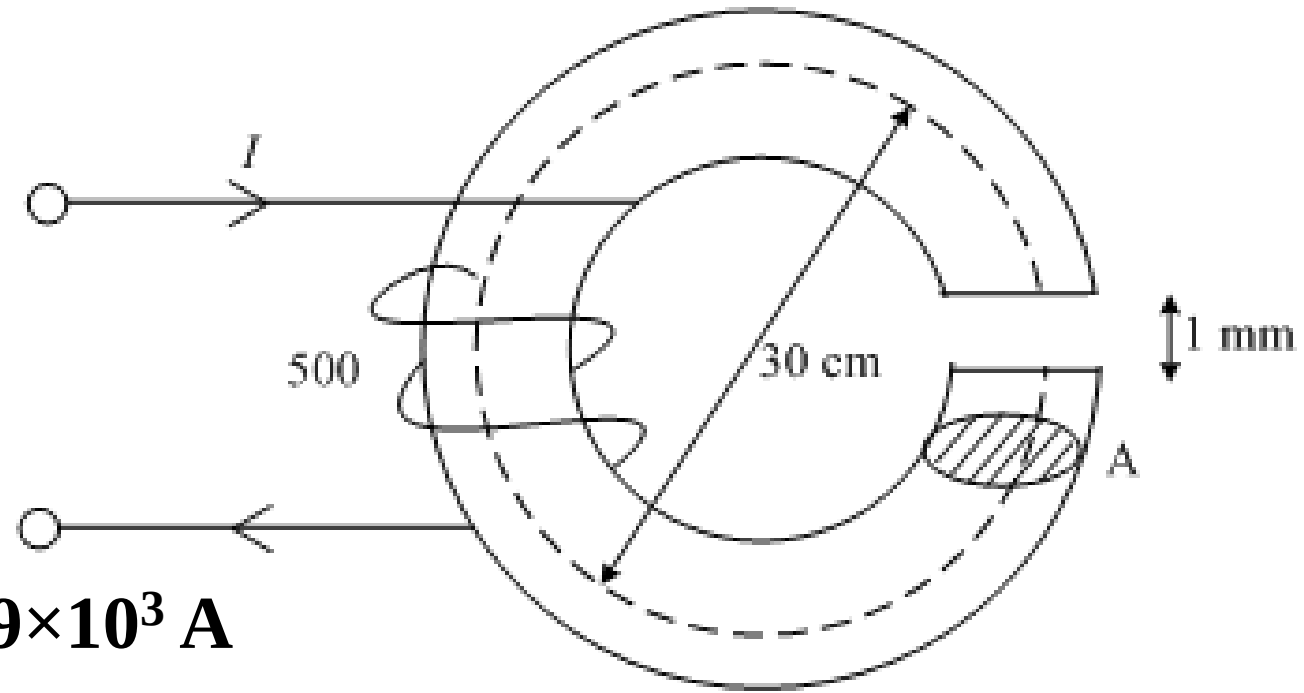
$$\Rightarrow \text{mmf across air gap } F_1 = H_1 l_1 = \mathbf{6.49 \times 10^3 \text{ A}}$$

Now consider iron ring:

$$H_2 = B / (\mu_r \mu_0) = 8.15 / (800 \times 4\pi \times 10^{-7}) \text{ A/m}$$
$$= 8.1 \times 10^3 \text{ A/m}$$

$$l_2 \sim 2\pi R = 2 \times 3.14 \times 0.15 \text{ m} = 0.94 \text{ m}$$

$$\Rightarrow \text{mmf across iron ring } F_2 = H_2 l_2 = \mathbf{7.63 \times 10^3 \text{ A}}$$



$$\text{Total mmf } F = F_1 + F_2$$
$$= \mathbf{1.41 \times 10^4 \text{ A}}$$

$$\Rightarrow \text{Current } I = F / N$$
$$= (1.41 \times 10^4) / 500 = \mathbf{28.24 \text{ A}}$$

SOLVED EXAMPLE

Q. An iron ring of mean circumference 50-cm has an air gap of 0.1-cm length and a winding of 300 turns. If the relative permeability of iron is 400 when a current of 1 ampere flows through the coil, find the flux density in the air gap.

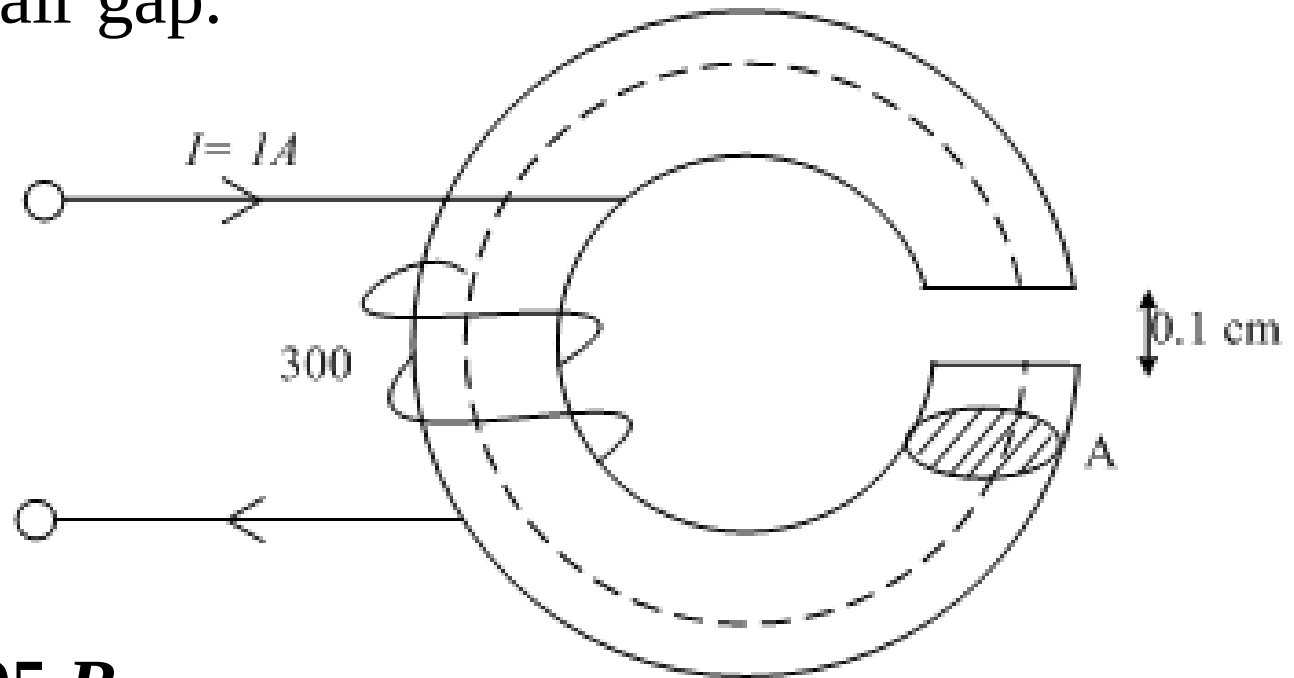
Ans: Let B be the flux density in the iron ring. Its value would also be the same in the air gap.

First consider iron ring:

$$\begin{aligned} H_1 &= B / (\mu_r \mu_0) = B / (400 \times 4\pi \times 10^{-7}) \\ &= 1989.44 B \end{aligned}$$

$$l_1 = 50 \text{ cm} = 0.5 \text{ m}$$

$$\Rightarrow \text{mmf across iron ring } F_1 = H_1 l_1 = \mathbf{995 B}$$



Now consider air gap:

$$H_2 = B / \mu_0 = B / (4\pi \times 10^{-7})$$
$$= 7.96 \times 10^5 B$$

$$l_2 = 0.1 \text{ cm} = 10^{-3} \text{ m}$$

$\Rightarrow$ *mmf* across air gap

$$F_2 = H_2 l_2 = \mathbf{796 B}$$

$$\text{Total } mmf F = F_1 + F_2$$

$$= (995 + 796)B = \mathbf{1791 B}$$

$$\text{But given total } mmf F = N.I = 300 \times 1 = \mathbf{300}$$

$$\Rightarrow \boxed{B} = 300/1791 = \boxed{\mathbf{0.1675 \text{ Wb/m}^2}}$$

